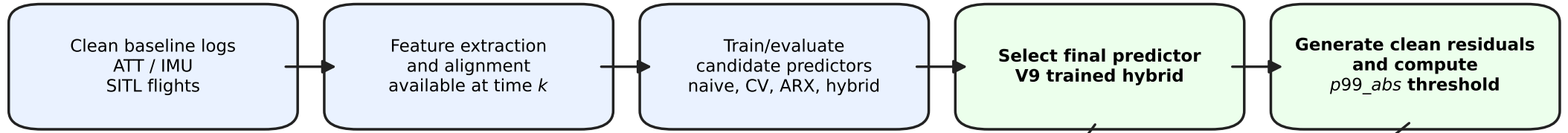


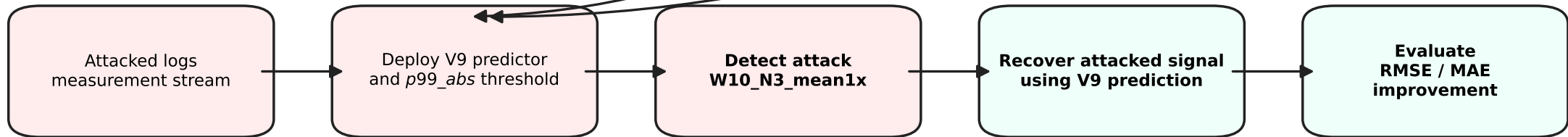
Figure 2. Training and Deployment Workflow

Offline training and calibration



selected V9 model $p99_abs$ threshold

Deployment on attacked logs



Clean logs are used to learn normal behavior and calibrate the detector; attacked logs are used for detection, recovery, and final evaluation.